

Day 04 – Git Merge & Merge Conflicts

Topics Covered

Git Merge
Fast-Forward Merge
Three-Way Merge
Merge Base
Merge Commit
Merge Conflicts
Conflict Markers
Conflict Resolution
ORT Merge Strategy
Pull Request Workflow
Merge Best Practices

Fast-Forward Merge

Moves the branch pointer forward. No merge commit is created.

Three-Way Merge

Used when both branches diverge. Git creates a merge commit with two parents.

Merge Base

The most recent common ancestor of two branches.

Merge Conflict

Occurs when the same lines are modified differently.

Conflict Resolution

Edit file → Remove markers → git add <file> → git commit

ORT

Git's default merge strategy for efficient automatic merging and conflict detection.

Production Workflow

Feature Branch → Commit → Push → Pull Request → Code Review → CI → Approval → Merge → Deploy

Commands

git merge <branch>
git status
git add <file>
git commit
git log --graph --oneline --all

Interview

Explain Git Merge, Fast-Forward Merge, Three-Way Merge, Merge Base, Merge Commit, Merge Conflicts, ORT.